3AL Hand-in 1 2024 Solutions

Due: Monday 26 February 11:00 pm

Please make sure all of your answers are clearly labelled.

1. Determine whether each statement is true or false. No explanation needed.

   (a) If H and K are subgroups of G and $H \subseteq K$, then H is a subgroup of K.

   (b) If H and K are subgroups of G, then $H \cap K$ is a subgroup of G.

   (c) Every homomorphism from a group to itself is an automorphism.

   (d) All homomorphisms preserve the orders of elements.

(a) True Since H is a subset of K and is a group in its own right, it must be a subgroup of K.

(b) True. Note that $\{1\} \subseteq H \cap K$. Now take $a, b \in H \cap K$, then ab^{-1} is in H and K as they are subgroups of G. So $H \cap K$ is a subgroup of G.

(c) False. An automorphism must be an isomorphism. For a nontrivial group G, the homomorphism $\varphi : G \to G$ which maps every element g to the identity 1_G is not an isomorphism.

(d) False. Same φ as above does not preserve the order of nonidentity elements.

[4 marks]

2. Prove or disprove: If $\varphi : G \to G'$ is a homomorphism and H is a subgroup of G, then $\varphi(H)$ is a subgroup of G'.

Since H is a subgroup of G, $1_G \in H$ and so $1_{G'} = \varphi(1_G) \in \varphi(H)$. Let $a', b' \in \varphi(H)$, then there exist $a, b \in H$ such that $\varphi(a) = a'$ and $\varphi(b) = b'$. Since H is a subgroup of G, we have $ab^{-1} \in H$. Thus $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = a'(b')^{-1} \in \varphi(H)$.

[4 marks]

3. Prove or disprove: If $\varphi : G \to G'$ is a homomorphism and $H \subseteq G$ is such that $\varphi(H)$ is a subgroup of G', then H is a subgroup of G.

e.g. Consider $\varphi : \mathbb{Z} \to \mathbb{Z}_2$ where $\varphi(n) = [0]$ for all $n \in \mathbb{Z}$. Then $\varphi(\{1\}) = \{[0]\} \leq \mathbb{Z}_2$ but $\{1\} \not\leq \mathbb{Z}$.

NOTE: Let $g \in G$. If $\varphi(g) \in \varphi(H)$, it doesn't necessarily mean $g \in H$. It's possible that multiple elements from G map to the same element of G'. i.e. g is not necessarily in H. It just means that there is some element $h \in H$ where $\varphi(g) = \varphi(h)$.

[4 marks]

Maximum mark = 10

Available marks = 12